Year 11 Mathematics

Shape and Space 6

10.1 Congruent Shapes

Congruent Shapes

STARTER · 5 MIN

Date: _____ **Title:** Congruent Shapes

LEARNING OBJECTIVES

- Recognise congruent shapes
- Use the conditions SSS, SAS, ASA and RHS
- Prove that two triangles are congruent

STARTER *Answer in your book. No calculators.*

1. Angles in a triangle add up to?
2. Find the third angle: 40° and 75°
3. Name a shape: 4 equal sides, no right angles
4. What stays the same in a rotation?

Starter — Answers

STARTER

Tick or correct in green pen

1. **180°**

2. $180 - 115 = \mathbf{65°}$

3. **Rhombus**

4. **Lengths and angles**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can recognise congruent shapes	☐
SILVER	I can use the conditions SSS, SAS, ASA and RHS	☐
GOLD	I can prove that two triangles are congruent	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Congruent Triangles

INTRODUCTION · 3 MIN

Congruent shapes are exactly the same shape and size. They may be rotated or reflected.

SSS

All three sides are equal.

Side, side, side

SAS

Two sides and the angle between them.

The angle must be included

ASA

Two angles and a matching side.

Angle, side, angle

RHS

Right angle, hypotenuse and one side.

Right-angled triangles only

Examples 1 and 2 — Conditions for Congruence

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint congruent if: SSS SAS ASA RHS

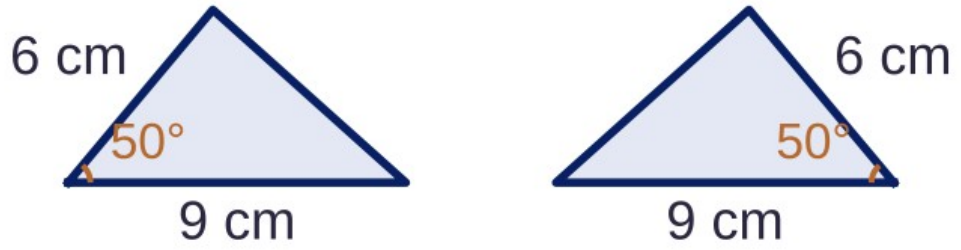
Example 1

Are these triangles congruent?
Give a reason.



Example 2

Are these triangles congruent?
Give a reason.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

All three sides match

Congruent (SSS)

Tip: check every side.

Example 2

Two sides and the angle between them

Congruent (SAS)

Tip: the angle is between the two sides.

Task 1

PRACTICE · 9 MIN

BRONZE

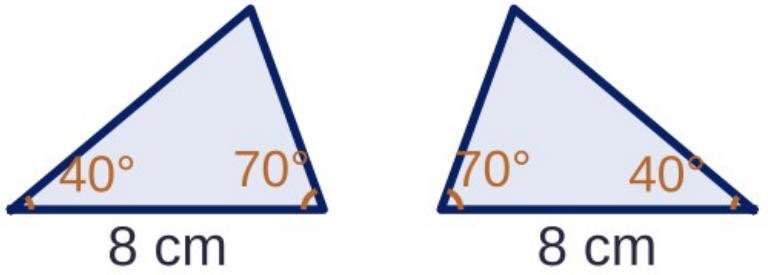
Congruent or not? Give the condition.

Hint congruent if: SSS SAS ASA RHS

1



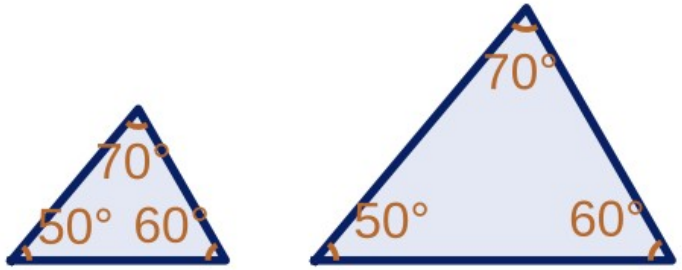
2



3



★



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **Congruent (SSS)**

2. **Congruent (ASA)**

3. **Congruent (RHS)**

★ **Not congruent : angles only**

Examples 3 and 4 — Using Congruent Triangles

EXAMPLES · 5 MIN

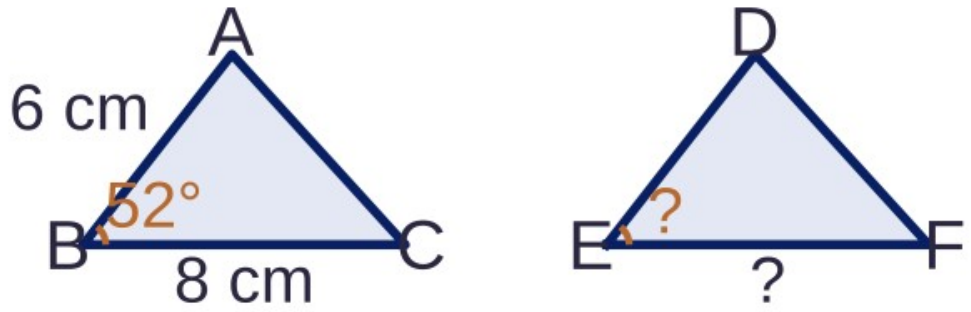
SILVER

Watch and copy into your book

Hint congruent if: SSS SAS ASA RHS

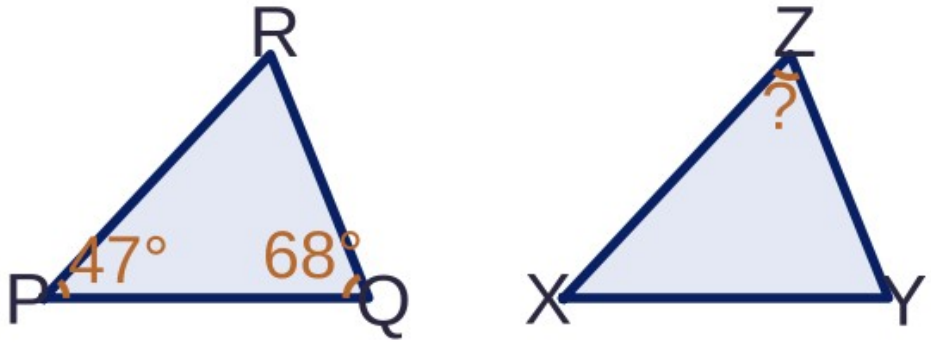
Example 3

ABC and DEF are congruent.
Find EF and angle E.



Example 4

PQR and XYZ are congruent.
Find angle Z.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

EF matches BC

EF = 8 cm

Angle E **= 52°**

Tip: match the letters in order.

Example 4

R = 180 − 47 − 68

= 65°

Z matches R

Angle Z **= 65°**

Tip: find the missing angle first.

Task 2

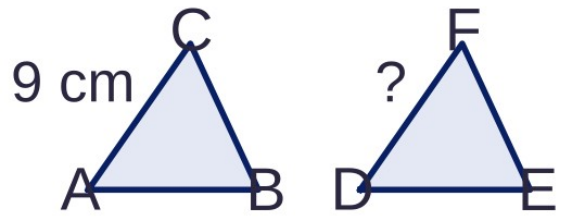
PRACTICE · 9 MIN

SILVER

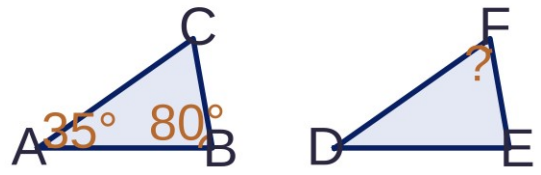
Answer in your book

Hint congruent if: SSS SAS ASA RHS

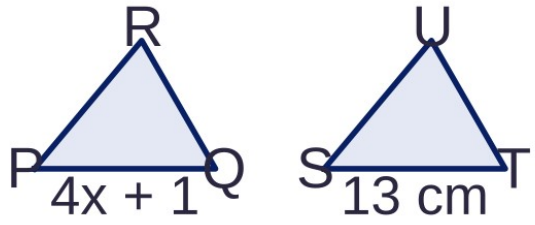
1 ABC ≅ DEF. Find DF.



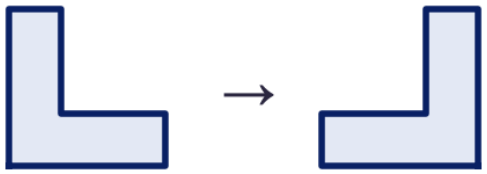
2 ABC ≅ DEF. Find angle F.



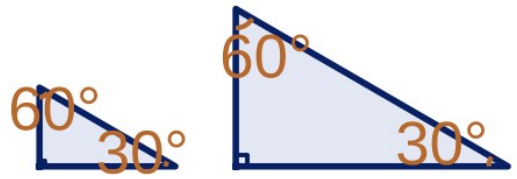
3 PQR ≅ STU. Find x.



4 Is the rotated shape congruent?



5 Must these be congruent?



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. DF = AC = **9 cm**

2. F = C = 180 − 115 = **65°**

3. $4x + 1 = 13, \quad x = \mathbf{3}$

4. **Yes : same size and shape**

5. **No : AAA does not fix the size**

Examples 5 and 6 — Proving Congruence

EXAMPLES · 5 MIN

GOLD

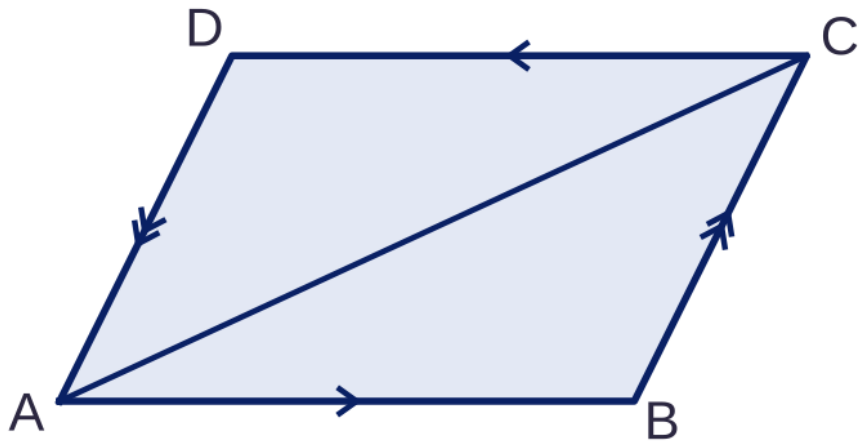
Exam-style: show all your working

Hint congruent if: SSS SAS ASA RHS

Example 5

(3 marks)

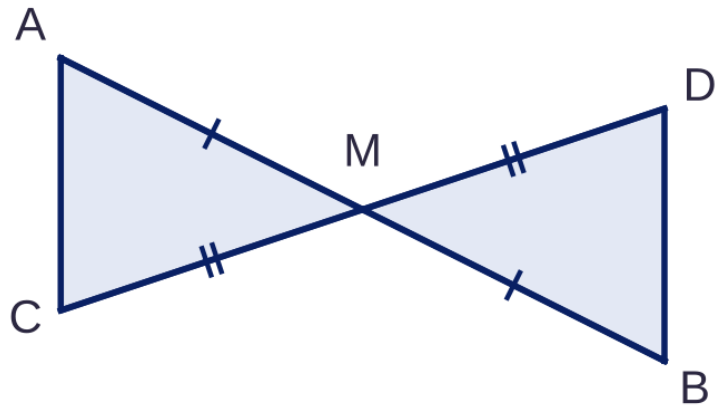
ABCD is a parallelogram. Prove ABC ≅ CDA.



Example 6

(3 marks)

M is the midpoint of AB and CD. Prove AMC ≅ BMD.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

AB = CD (opposite sides)

BC = DA (opposite sides)

AC is a common side

Congruent (SSS)

Tip: give a reason for every pair.

Example 6

AM = MB (midpoint)

CM = MD (midpoint)

AMC = BMD (vertically opposite)

Congruent (SAS)

Tip: look for vertically opposite angles.

Task 3

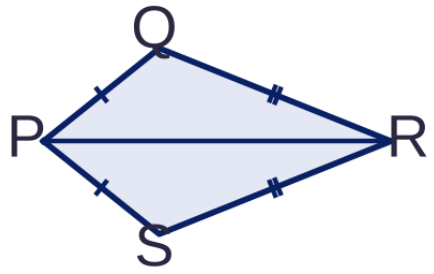
PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

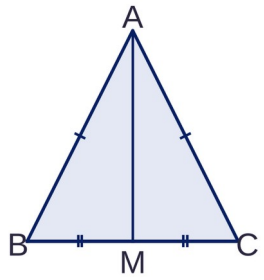
Hint congruent if: SSS SAS ASA RHS

1 Prove PQR ≅ PSR.



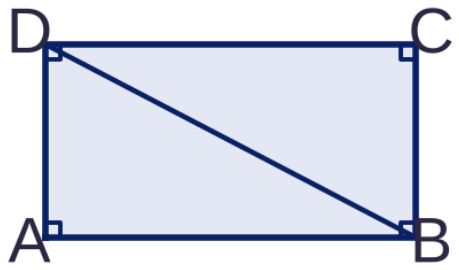
(3 marks)

2 Prove ABM ≅ ACM.



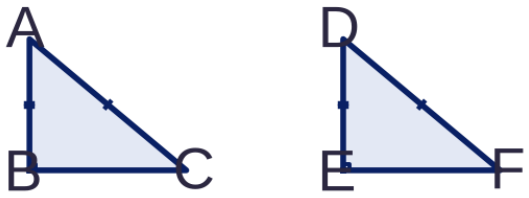
(3 marks)

3 Prove ABD ≅ CDB.



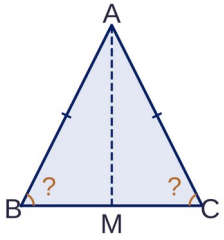
(3 marks)

4 Which condition proves ABC ≅ DEF?



(1 mark)

5 Prove that angle B = angle C.



(3 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. PQ = PS, QR = SR, PR common, **SSS**
2. AB = AC, BM = CM, AM common, **SSS**
3. AB = CD, AD = CB, BD common, **SSS**
4. **RHS**
5. ABM ≅ ACM (SSS), so angle B = **angle C**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Two triangles have all three sides equal. Which condition is this?

SILVER ABC ≅ DEF and angle C = 72°. Write down angle F.

GOLD Why is AAA not a condition for congruence?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **SSS**

SILVER F = C = **72°**

GOLD **The size can be different**

Exam Practice — Conditions for Congruence

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint congruent if: SSS SAS ASA RHS

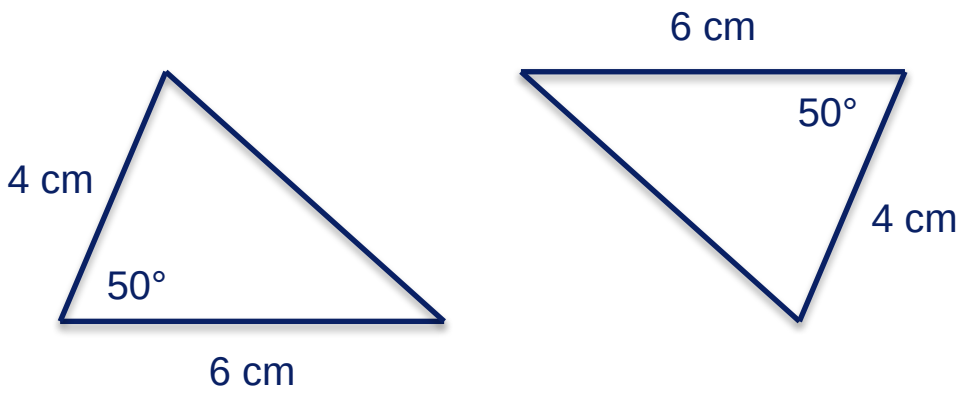
Question 1

(2 marks)

Are these triangles congruent?

Give a reason.

Diagram NOT accurately drawn



.....

Question 2

(2 marks)

Triangles ABC and DEF both have a right angle.

The hypotenuses are both 10 cm.

BC = EF = 6 cm.

Are the triangles congruent? Give a reason.

.....

Exam Practice — Using Congruent Triangles

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint congruent if: SSS SAS ASA RHS

Question 3

(3 marks)

Triangle ABC is congruent to triangle DEF.

AB = 8 cm, BC = 11 cm, angle A = 64° and angle B = 47°.

(a) Write down the length of EF. (1)

..... cm

(b) Work out angle F. (2)

..... °

Question 4

(2 marks)

Two triangles both have angles of 30°, 70° and 80°.

Must they be congruent? Explain.

.....

Exam Practice — Proving Congruence

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint congruent if: SSS SAS ASA RHS

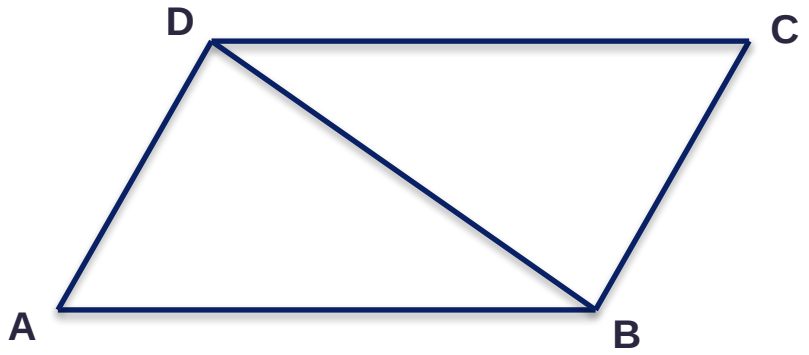
Question 5

(3 marks)

ABCD is a parallelogram.

Prove that triangles ABD and CDB are congruent.

Diagram NOT accurately drawn



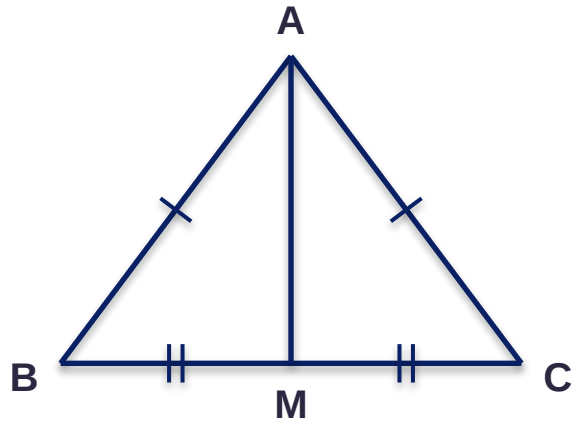
Question 6

(3 marks)

AB = AC and M is the midpoint of BC.

Prove that triangles ABM and ACM are congruent.

Diagram NOT accurately drawn



Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

Yes B1

SAS: two sides and the angle B1

between them are equal

Tip: 50° is between 6 cm and 4 cm.

Question 2

Yes B1

RHS: right angle, hypotenuse B1

and one side

Tip: RHS only works for right-angled triangles.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) EF = BC = 11 cm — B1

(b) F = C = 180 − 64 − 47 — M1

= 69° — A1

Tip: match the letters in order: A ↔ D, B ↔ E, C ↔ F.

Question 4

No — B1

same angles: only similar, — B1

one can be bigger

Tip: AAA is not a condition for congruence.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

AB = CD (opposite sides)	B1
AD = CB (opposite sides)	B1
BD is common, so SSS	B1

Tip: give a reason for every pair.

Question 6

AB = AC (given)	B1
BM = CM (M is the midpoint)	B1
AM is common, so SSS	B1

Tip: end with the condition.